

Building Molecules Atom by Atom

PhET Build a Molecule · PhET

Years 7-8

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/build-a-molecule>

Curriculum links: AC9S7U05 · AC9S8U06 · bonding

Learning intentions

- I can build simple molecules from individual atoms and read their chemical formula.
- I can describe the difference between an atom, a molecule and a compound.
- I can explain, using particle ideas, why atoms join together to form molecules.

Getting started

Open PhET Build a Molecule. Start on the first screen, where you can drag atoms from the bucket at the bottom into the workspace to join them together. Try building water (H₂O) first by dragging two hydrogen atoms and one oxygen atom together, just to see how the simulation shows a completed molecule.

Tasks

1. Drag one carbon atom and two oxygen atoms into the workspace and join them together. What is the name and chemical formula of the molecule you have built?

2. Try dragging just one hydrogen atom and one oxygen atom together. Does the simulation let you 'complete' a molecule? What does this tell you about how many atoms some molecules need?

3. Use the collection box to select four different target molecules. Build each one and record what atoms you used and the formula the simulation gives you.

Molecule built	Atoms used	Chemical formula

4. Choose one molecule you built that contains two of the same atom (for example O₂ or H₂O). Write out its formula and mark the repeated atom's symbol.

5. Explain, in your own words, the difference between an atom and a molecule, using two examples from the molecules you built.

6. Build methane (CH₄) if it's available as a target, or another molecule with several atoms attached to one central atom. Describe how the atoms are arranged around the central atom.

7. Atoms in a molecule are held together by forces called bonds. Based on what you observed when dragging atoms together (did they snap together or stay separate?), explain in your own words what you think happens between atoms when they bond.

Extension (optional)

Table salt (sodium chloride) and sugar (a much larger molecule) are both compounds made of more than one type of atom. Using the idea of 'formula' from this simulation, explain why a chemical formula like NaCl or C₆H₁₂O₆ is a useful shorthand for scientists.
